JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SATURDAY 04th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

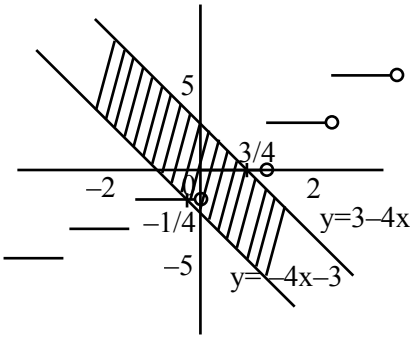
1. Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha+\beta)$ is equal to :

(1) 6 (2) 8

(3) 9 (4) 4

Sol. $-1 \le \frac{4x+2[x]}{3} \le 1$

$-3 - 4x \le 2[x] \le 3 - 4x$



$x \in \left[\frac{-1}{4}, \frac{3}{4}\right]$

$12(\alpha+\beta) = 6$

2. If the set of all solutions of $|x^2 + x - 9| = |x| + |x^2 - 9|$ is $[\alpha, \beta] \cup [\gamma, \infty)$, then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to :

(1) 9 (2) 18

(3) 36 (4) 72

Sol. $|a| + |b| = |a + b|$

$\Rightarrow a \cdot b \ge 0$

$\Rightarrow x(x^2 - 9) \ge 0$

$x \in [-3, 0] \cup [3, \infty]$

$\alpha^2 + \beta^2 + \gamma^2 = 18$

3. Let z be a complex number such that $|z + 2| = |z - 2|$ and $\arg\left(\frac{z+3}{z-i}\right) = \frac{\pi}{4}$. Then $|z|^2$ is equal to :

(1) 9 (2) 4

(3) 5 (4) 1

Sol. $|z + 2| = |z - 2|$

$\Rightarrow z = 0 + iy \quad y \in R$

$\arg\left(\frac{3+iy}{iy-i}\right) = \frac{\pi}{4}$

$\Rightarrow \arg\left(\frac{y}{y-1} + \frac{3i}{1-y}\right) = \frac{\pi}{4}$

$\Rightarrow \frac{y}{y-1} = \frac{3}{1-y} > 0$

$\Rightarrow z = -3i$

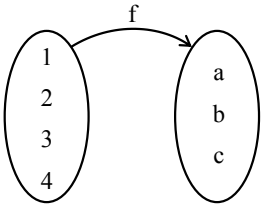
$|z|^2 = 9$

4. The number of functions $f : \{1, 2, 3, 4\} \to \{a, b, c\}$, which are not onto, is :

(1) 48 (2) 45

(3) 51 (4) 35

Sol.



No. of into function

$= {}^3C_1\, 2^4 - {}^3C_2\, 1 + 0$

$= 3 \times 16 - 3 \times 1$

$= 45$

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5. Let

$$S=\left\{A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}: a,b,c,d\in\{0,1,2,3,4\} \text{ and } A^2-4A+3I=0\right\}$$

be a set of 2 × 2 matrices. Then the number of matrices in S, for which the sum of the diagonal elements is equal to 4, is:

(1) 20 (2) 17

(3) 21 (4) 19

Sol. a, b, c, d ∈ {0, 1, 2, 3, 4}

$a + d = 4 \quad ad - bc = 3$

Case-I $a = 0 \quad d = 4$

$0 - bc = 3$

No solution

Case-II $a = 1 \quad d = 3$

$3 - bc = 3$

$bc = 0 \quad \Rightarrow$ 9 ways

Case-III $a = 2 \quad d = 2$

$4 - bc = 3$

$bc = 1 \quad \Rightarrow$ 1 ways

Case-IV $a = 3 \quad d = 1$

$bc = 0 \quad \Rightarrow$ 9 way

Case-V $a = -4 \quad d = 0 \Rightarrow$ No Solution

Total 19 matrix

6. Let $A=\begin{bmatrix}1 & 1 & 2\\ -2 & 0 & 1\\ 1 & 3 & 5\end{bmatrix}$. Then the sum of all elements of the matrix adj(adj $(2(\text{adj}A)^{-1})$) is equal to :

(1) 3 (2) 4

(3) –4 (4) –3

Sol. $|A| = -4$

$$\text{As } (\text{adj}A)^{-1}=\frac{A}{|A|}=\frac{-A}{4}$$

$$B=2(\text{adj}A)^{-1}=\frac{-A}{2}$$

$$|B|=\left|\frac{-A}{2}\right|$$

$$=\frac{-1}{8}|A|=\frac{1}{2}$$

$$\text{adj}(\text{adj}B)=|B|^{n-2}B=|B|B$$

$$=\frac{B}{2}=\frac{-A}{4}$$

$$\text{Sum of element}=\frac{-12}{4}=-3$$

7. The first term of an A.P. of 30 non-negative terms is $\frac{10}{3}$. If the sum of this A.P. is the cube of its last term, then its common difference is :

(1) $\frac{5}{87}$ (2) $\frac{25}{83}$

(3) $\frac{15}{29}$ (4) $\frac{5}{29}$

Sol. $a=\frac{10}{3}: S_{30}=(T_{30})^3$

$$\frac{30}{2}\left[2\times\frac{10}{3}+29d\right]=\left[\frac{10}{3}+29d\right]^3$$

$$15\left(\frac{20}{3}+29d\right)=\left(\frac{10}{3}+29d\right)^3$$

$$15\left(\frac{10}{3}+\frac{10}{3}+29d\right)=\left(\frac{10}{3}+29d\right)^3$$

By observation $\frac{10}{3}+29d=5$

$$29d=5-\frac{10}{3}=\frac{5}{3}$$

$$d=\frac{5}{87}$$

8. The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:

(1) 5040 (2) 3050

(3) 3410 (4) 4320

Sol. No. of ways to arrange 7 students = 7!

Cases where all girls are together

$B_1B_2B_3B_4\ \boxed{G_1G_2G_3}$ = 5! 3!

= 7! – 6!

= 6 × 6!

= 6 × 720

= 4320

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9. Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \ldots + (1+x)^{99} + (1+kx)^{100}$, $x \neq 0$, is $\left(43n + \frac{101}{4}\right)\left({}^{100}C_3\right)$ for some $n \in \mathbb{N}$, be p. Then the value of $p + n$ is:

(1) 10 (2) 11

(3) 12 (4) 13

Sol. ${}^3C_3 + {}^4C_3 + {}^5C_3 + \ldots + {}^{99}C_3 + {}^{100}C_3K^3 = \left(43n + \frac{101}{4}\right){}^{100}C_3$

$\Rightarrow {}^4C_4 + {}^4C_3 + {}^5C_3 + \ldots + {}^{99}C_3 + {}^{100}C_3K^3 = \left(43n + \frac{101}{4}\right){}^{100}C_3$

$= {}^{100}C_4 + {}^{100}C_3K^3 = \left(43n + \frac{101}{4}\right){}^{100}C_3$

$\frac{{}^{100}C_4}{{}^{100}C_3} + K^3 = 43n + \frac{101}{4}$

$\frac{97}{4} + K^3 = 43n + \frac{101}{4}$

$K^3 = 43n + 1$

$n = 5$

$K^3 = 216$

$K = 6 = P$

$P + n = 11$

10. Suppose that the mean and median of the non-negative numbers 21, 8, 17, a, 51, 103, b, 13, 67, (a > b), are 40 and 21, respectively. If the mean deviation about the median is 26, then 2a is equal to :

(1) 109 (2) 117

(3) 161 (4) 131

Sol. 8, 13, b, 17, 21, 51, a, 67, 103

$\text{mean} = 40 = \frac{280 + a + b}{9}$

$a + b = 80$

mean deviation about median

$= \frac{13 + 8 + 4 + 30 + 46 + 82 + (a - 21) + (21 - b)}{9} = 26$

$a - b = 51$

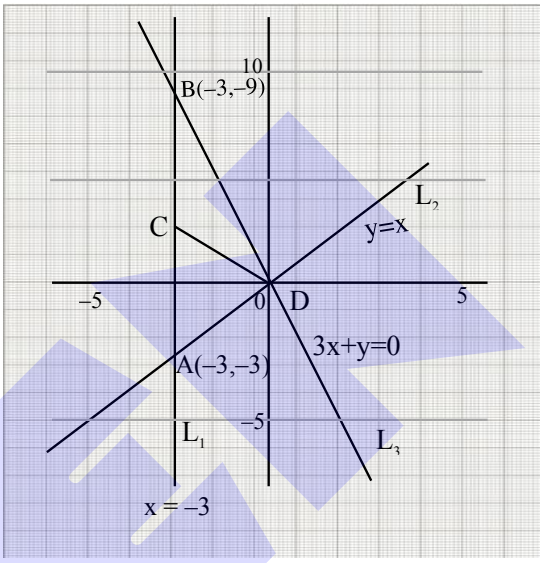
$\Rightarrow 2a = 131$

11. Let the line $L_1 : x + 3 = 0$ intersect the lines $L_2 : x - y = 0$ and $L_3 : 3x + y = 0$ at the points A and B, respectively. Let the bisector of the obtuse angle between the lines L_2 and L_3 intersect the line L_1 at the point C. Then $BC^2 : AC^2$ is equal to :

(1) 5 : 1 (2) 1 : 5

(3) 2 : 3 (4) 3 : 2

Sol.



$\frac{BC^2}{AC^2} = \left(\frac{OB}{OA}\right)^2 = \frac{9 + 81}{9 + 9}$

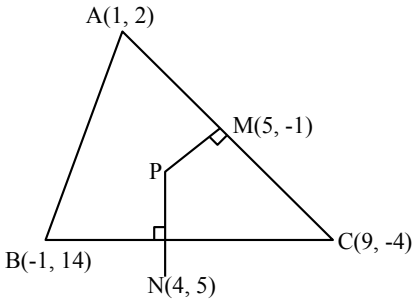
$= 5$

12. Let the vertex A of a triangle ABC be (1, 2), and the mid-point of the side AB be (5, –1). If the centroid of this triangle is (3, 4) and its circumcenter is (α, β), then $21(\alpha + \beta)$ is equal to:

(1) 309 (2) 403

(3) 497 (4) 524

Sol.



equation of PM : $4x - 3y = 23$

equation of PM : $5x - 9y = -25$

$P\left(\frac{94}{7}, \frac{215}{21}\right)$

$21(\alpha + \beta) = 497$

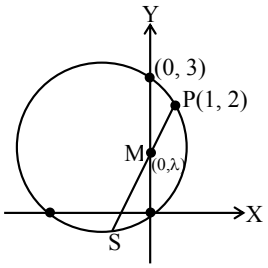
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13. Suppose the two chords, drawn from the point (1, 2) on the circle $x^2 + y^2 + x - 3y = 0$ are bisected by the y-axis. If the other ends of these chords are R and S, and the mid point of the line segment RS is (α, β), then $6(\alpha + \beta)$ is equal to :

(1) 1 (2) 3

(3) 4 (4) 6

Sol.



chord PS bisected by y –axis

at point M(0, λ)

$\therefore$ S $(-1, 2\lambda - 2)$

Which satisfy equation of circle

$\Rightarrow (-1)^2 + (2\lambda - 2)^2 - 1 - 3(2\lambda - 2) = 0$

$\Rightarrow \lambda = 1 \quad \& \quad \lambda = \frac{5}{2}$

Therefore two such chord PS & PR will exist

When $\lambda = 1 \Rightarrow$ S(–1, 0)

When $\lambda = \frac{5}{2}$ R(–1,3)

$\therefore$ Mid point of R & S will be $\left(-1, \frac{3}{2}\right)$

$\therefore \alpha = -1, \ \beta = \frac{3}{2}$

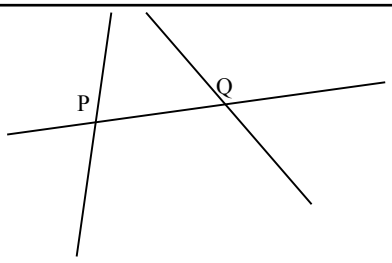
$\therefore 6(\alpha + \beta) = 3$

14. A line with direction ratios 1, –1, 2 intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q, respectively. If the length of the line segment PQ is α, then $225\alpha^2$ is equal to:

(1) 1024 (2) 1014

(3) 1104 (4) 1204

Sol.



P(2λ, 3λ, 3λ – 1) and Q (–μ – 1, μ + 2, 4μ)

D.R of PQ 2λ + μ + 1, 3λ – μ – 2, 3λ – 4μ – 1,

$\frac{2\lambda + \mu + 1}{1} = \frac{3\lambda - \mu - 2}{-1} = \frac{3\lambda - 4\mu - 1}{2}$

Solving $\lambda = \frac{1}{5}$ and $\mu = \frac{-8}{15}$

$P\left(\frac{2}{5}, \frac{3}{5}, \frac{-2}{5}\right) \& Q\left(\frac{-7}{15}, \frac{22}{15}, \frac{-32}{15}\right)$

$PQ^2 = \left(\frac{13}{15}\right)^2 + \left(\frac{13}{15}\right)^2 + \left(\frac{26}{15}\right)^2$

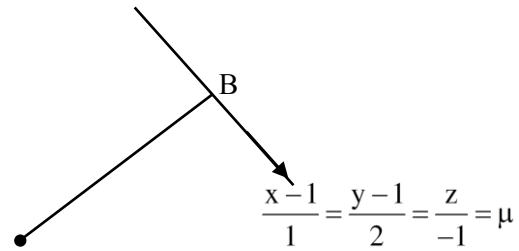
$\alpha^2 = \frac{1014}{225} \ \therefore 225\alpha^2 = 1014$

15. The square of the distance of the point (–2, –8, 6) from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:

(1) 3 (2) 6

(3) 8 (4) 12

Sol.



A(–2, –8, 6)

DR or AB is 1, –1, 2

$\therefore$ Line AB is $\frac{x+2}{1} = \frac{y+8}{-1} = \frac{z-6}{2} = \lambda$

B is (λ–2, –λ–8, 2λ+6)

B is (μ+1, 2μ+1, –μ)

Solving point B is (–3, –7, 4)

$AB = \sqrt{(-1)^2 + (1)^2 + (2)^2} = \sqrt{6}$

$AB^2 = 6$

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16. If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is equal to :

(1) 3 (2) –1

(3) 1 (4) 2

Sol. $y = \tan^{-1}\left(\frac{\frac{3}{4} - \tan x}{1 + \frac{3}{4}\tan x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$

For $2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$

put $x = \sin\theta$

$\therefore\ 2\tan^{-1}\left(\frac{\sin\theta}{1+\cos\theta}\right) = \theta = \sin^{-1}x$

Hence

$y = \tan^{-1}\left(\frac{3}{4}\right) - \tan^{-1}(\tan x) + \sin^{-1}x$

$\frac{dy}{dx} = -1 + \frac{1}{\sqrt{1-x^2}}$

$x = \frac{\sqrt{3}}{2}$

$\frac{dy}{dx} = -1 + 2 = 1$

17. Let f be a real polynomial of degree n such that $f(x) = f'(x)\, f''(x)$, for all $x \in \mathbb{R}$. If $f(0) = 0$, then $36\left(f'(2) + f''(2) + \int_0^2 f(x)dx\right)$ is equal to :

(1) 42 (2) 46

(3) 56 (4) 66

Sol. Let degree of polynomial be 'n'

$n = n - 1 + n - 2 \Rightarrow n = 3$

$f(x) = ax^3 + bx^2 + cx$

$(ax^3 + bx^2 + cx) = (3ax^2 + 2bx + c)(6ax + 2b)$

$x^3 : a = 18a^2 \Rightarrow a = \frac{1}{18}$

$x^2 : b = 6ab + 12ab$

$x : c = 4b^2 + 6ac \Rightarrow \frac{2C}{3} = 4b^2 \Rightarrow b^2 = \frac{c}{6}$

const : $bc = 0 \Rightarrow b = c = 0$

$\int_0^2 f(x)dx + f'(2) + f''(2)$

$\int_0^2 ax^3dx + 12a + 12a = 4a + 24a = 28a = \frac{28}{18} = \frac{14}{9}$

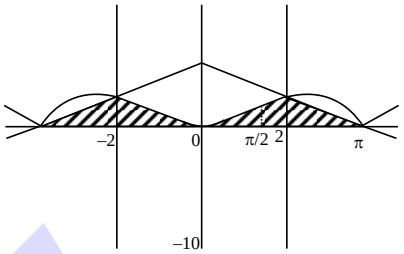
$36\left(\frac{14}{9}\right) = 56$

18. The area of the region $\{(x,y): y \le \pi - |x|,\ y \le |x\sin x|,\ y \ge 0\}$ is :

(1) $1 + \frac{\pi^2}{8}$ (2) $2 + \frac{\pi^2}{4}$

(3) $\frac{\pi^2}{8} - 1$ (4) $4 + \frac{\pi^2}{2}$

Sol.



Area ; $2\left(\int_0^{\pi/2} x\sin x dx + \int_{\pi/2}^{\pi} (\pi - x)dx\right)$

$= 2(-x\cos x + \sin x)_0^{\pi/2} + 2\left(\frac{(\pi-x)^2}{2}\Big|_{\pi}^{\pi/2}\right)$

$= 2 + \frac{\pi^2}{4}$

19. Let $\int_{-2}^{2} (|\sin x| + [x\sin x])dx = 2(3 - \cos 2) + \beta$, where $[\cdot]$ is the greatest integer function . Then $\beta \sin\left(\frac{\beta}{2}\right)$ equals.

(1) 1 (2) 2

(3) 4 (4) 8

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Sol. $\int_{-2}^{2}(|\sin x|+[x\sin x])dx$

$\Rightarrow 2\int_{0}^{2}(|\sin x|+[x\sin x])dx$

$\Rightarrow 2\left[\int_{0}^{2}\sin x dx+\int_{0}^{2}[x\sin x]dx\right]$

$2\left([-\cos x]_0^2+\int_0^{\alpha}0dx+\int_{\alpha}^{2}2dx\right)$

$2(3-\cos 2-2-\alpha)$

$2(3-\cos 2)-2\alpha$

where α is Root of $x\sin x = 1$

Now comparing $\beta = -2\alpha$

$$\alpha = -\frac{\beta}{2}$$

$$-\frac{\beta}{2}\sin\left(\frac{-\beta}{2}\right)=1$$

$$\beta\sin\left(\frac{\beta}{2}\right)=2$$

20. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = (1 + x + x^2)(1 - y + y^2)$, $y(0) = \frac{1}{2}$. Then $(2y(1)-1)$ is equal to

(1) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$ (2) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

(3) $\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$ (4) $\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

$$\frac{dy}{\left(y-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}=(x^2+x+1)dx$$

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{y-\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)=\frac{x^3}{3}+\frac{x^3}{2}+x+C$$

$$\frac{2}{\sqrt{3}\sqrt{3}}=\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{x^3}{3}+\frac{x^3}{2}+x+C$$

$$y(0)=\frac{1}{2}\Rightarrow 0=0+0+0+C\Rightarrow C=0$$

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{x^3}{3}+\frac{x^2}{2}+x$$

put x = 1

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{1}{3}+\frac{1}{2}+1=\frac{2+3+6}{6}=\frac{11}{6}$$

$$\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{11\sqrt{3}}{12}$$

$$\frac{2y-1}{\sqrt{3}}=\tan\left(\frac{11\sqrt{3}}{12}\right)$$

$$\Rightarrow 2y(1)-1=\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$$

SECTION-B

21. A coin is tossed 8 times. If the probability that exactly 4 heads appear in the first six tosses and exactly 3 heads appear in the last five tosses is p, then 96p is equal to _____.

Sol. $x_1x_2x_3x_4x_5x_6x_7x_8$

Exactly 4 heads in first six Tosses.

$(x_1x_2x_3x_4x_5x_6)$

Exactly 3 heads in last 5 Tosses.

i.e. $(x_4\,x_5\,x_6\,x_7\,x_8)$

Common tosses $(x_4x_5x_6)$

Let k be number of heads on $(x_4x_5x_6)$

$k = 1 \;\therefore$ ways to overlap $= {}^3C_1$

${}^3C_3 \times {}^3C_1 \times {}^2C_2 = 3$ ways

$k = 2$

${}^3C_2 \times {}^3C_2 \times {}^2C_1 = 18$

$k = 3$

${}^3C_1 \times 1 = 3$

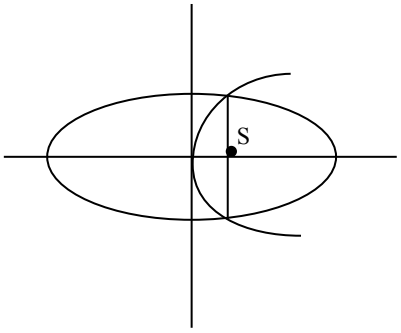
Total ways $24\left(\frac{1}{2}\right)^8$

$$96\times\frac{24}{28}=\frac{6\times 24}{16}=9$$

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22. Consider the parabola P : $y^2 = 4kx$ and the ellipse E : $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Let the line segment joining the points of intersection of P and E, be their latus rectums. If the eccentricity of E is e, then $e^2 + 2\sqrt{2}$ is equal to _____.

Sol.



Focus of parabola (k, 0)
Focus of Ellipse (ae, 0)
$k = ae$ .....(1)

Also $4k = 2\frac{b^2}{a}$ .....(2)

$4ae = \frac{2b^2}{a}$

$\Rightarrow 4a^2e = 2b^2$

$4a^2e = 2a^2(1 - e^2)$

$2e = 1 - e^2$

$e^2 + 2e + 1 = 2$

$(e + 1)^2 = 2$

$e + 1 = \sqrt{2} \quad \therefore e = \sqrt{2} - 1$

$e^2 + 2\sqrt{2} = 3$

23. If $A = \frac{\sin 3°}{\cos 9°}+\frac{\sin 9°}{\cos 27°}+\frac{\sin 27°}{\cos 81°}$ and B = tan81° – tan3°, then $\frac{B}{A}$ is equal to _____.

Sol. Consider $E = \frac{\sin\theta}{\cos 3\theta} \Rightarrow E = \frac{2\sin\theta\cos\theta}{2\cos 3\theta\cos\theta}$

$\Rightarrow E = \frac{\sin 2\theta}{2\cos 3\theta\cos\theta} \Rightarrow E = \frac{\sin(30-\theta)}{2\cos 3\theta\cos\theta}$

$\Rightarrow E = \frac{1}{2}[\tan 3\theta - \tan\theta]$

$A = \frac{1}{2}[\tan 9° - \tan 3° + \tan 27° - \tan 9° + \tan 81° - \tan 27°]$

$\therefore A = \frac{1}{2}[\tan 81° - \tan 3°]$

$\therefore \frac{B}{A} = 2$

24. Let $\vec{a}_k = (\tan\theta_k)\hat{i} + \hat{j}$ and $\vec{b}_k = \hat{i} - (\cos\theta_k)\hat{j}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n+1}$, for some $n \in \mathbb{N}$, n > 5. Then the value of $\frac{\sum_{k=1}^{n}|\vec{a}_k|^2}{\sum_{k=1}^{n}|\vec{b}_k|^2}$ is _____.

Sol. $|a_k|^2 = 1 + \tan^2\theta_k = \sec^2\theta_k$

$|b_k|^2 = 1 + \cot^2\theta_k = \text{cosec}^2\theta_k$

Since, $\cot\theta - \tan\theta = 2\cot 2\theta$

$\therefore -\text{cosec}^2\theta - \sec^2\theta = -4\,\text{cosec}^2 2\theta$

$\sec^2\theta = 4\text{cosec}^2 2\theta - \text{cosec}^2\theta$

$\Sigma \sec^2\theta_k = 4\Sigma\,\text{cosec}^2 2\theta_k - \Sigma\,\text{cosec}^2\theta_k$

Now

$\Sigma\,\text{cosec}^2 2\theta_k = \Sigma\,\text{cosec}^2\theta_k$

since

$\sum_{k=1}^{n}\text{cosec}^2\frac{2^k\pi}{2^n+1} = \sum_{k=1}^{n}\text{cosec}^2\frac{2^{k-1}\pi}{2^n+1}$

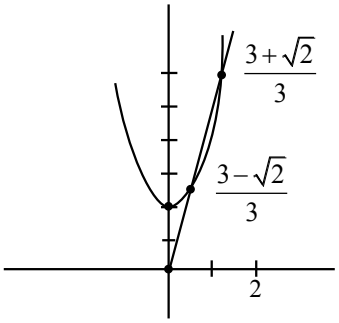
because $\text{cosec}\frac{2^n\pi}{2^n+1} = \text{cosec}\frac{\pi}{2^n+1}$

$\therefore \Sigma \sec^2\theta_k = 3\Sigma\,\text{cosec}^2\theta_k$

$\frac{\sum\sec^2\theta_k}{\sum\text{cosec}^2\theta_k} = 3$

25. The number of points, at which the fucntion $f(x) = \max\{6x, 2 + 3x^2\} + |x - 1|\cos\left|x^2 - \frac{1}{4}\right|, x \in (-\pi, \pi)$, is not differentiable, is _____.

Sol. $2 + 3x^2 = 6x \Rightarrow 3x^2 - 6x + 2 = 0$



$x = \frac{6 \pm \sqrt{12}}{6} = \frac{3 \pm \sqrt{2}}{3}$

$|x-1|\cos\left|x^2 - \frac{1}{4}\right|$ is differentiable at x = 1

Total number of points where $f(x)$ is non-Differentiable is 3

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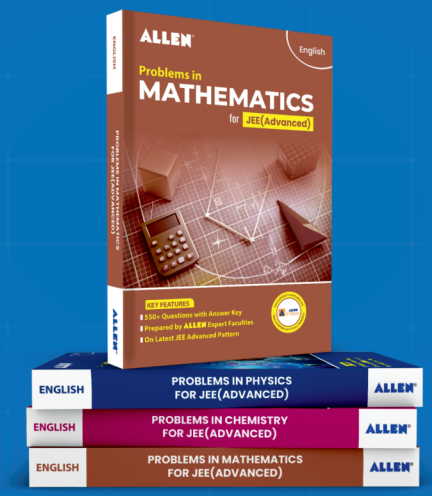


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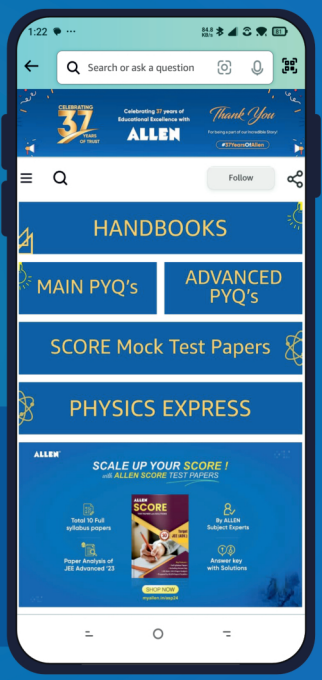
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